NAME(S):

Multi-branch if statements

Begin the following as paired-programming in class, but be sure to individually finish any problems you don't get to in order to be prepared for the practical exam

- hand in written answers of 0.25 engagement points (check website for last date accepted for credit)

Reminders for Pair Programming

- Instructions:
  - driver (has the keyboard and types) and navigator (guides the driver)
  - You should be talking to each other constantly!
  - You should be able to swap roles at any moment.
  - Swap roles every 5 to 10 minutes.

Learning goals / relevent topics

- Conditionals (Runestone Ch. 4)
- Comparison Operators (Runestone Ch. 2)
- Reeborg (https://reeborg.ca/docs/en/reference/index.html)

All of our work with conditionals thus far has only required `if` and `else` statements (where we only check whether a single condition is `True` or `False`). But in class, we've discussed elif statements, which can be used to distinguish between several conditional outcomes. For example, we can check whether a number `x` is `> 0`, `< 0`, or otherwise (`else`) equal to `0`.

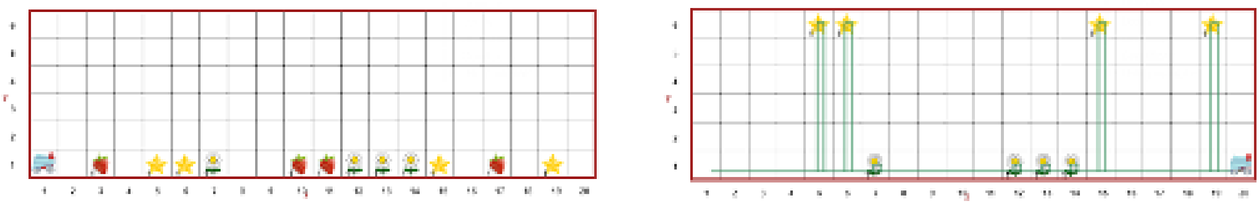
In today's exercise, we're going to utilize `elif` statements to account for more than two possible outcomes. Recall that "elif" is a contraction of "else" and "if" and that many other programming languages require the use of `else if` instead of the contracted form. You can have as many `elif` branches as you need.

Walk Through Field

1. Download the Walk Through Field folder (or its contents) from the course website, which should contain 2 json files, which each define a single world for Reeborg.
2. Import both worlds into Reeborg and see what each world looks like.
3. Reeborg's goal is to walk along the bottom of the world and:
   a. pick all strawberries
   b. put any fallen stars back into the sky
   c. perform a dance (turn 360 degrees) for each daisy

Below are before and after pictures for Field 0.

4. Write a program that accomplishes these goals for both worlds.



(a) Field Before

(b) Field After

Hint: the `object_here()` function can be given a string argument to check if a specific type of object is present (e.g. `object_here("token")`).

Logical Operators

In Python, logical operators allow you to combine two conditions into a single, more complex condition. Python has three logical operators: and, or, and not.

`<condition> and <condition>` is `True` if both conditions are `True`. `<condition> or <condition>` is `True` if at least one of the conditions is `True`. `not <condition>` is `True` the condition is `False`.

1. Download the Harvests folder from the course website and import each json file into Reeborg.
2. Find `harvest1.py` from the folder and copy the code into Reeborg.
3. Open the world Harvest 1 (from Reeborg's "introduction" worlds).
4. Run the code from `harvest1.py` and make sure you understand what it's doing.
5. Choose the world `harvest1a.json` (from the folder). You should notice that there are some holes in the field. Adapt the provided code so that Reeborg will harvest all of the carrots.

Note: the remaining portions also use worlds based on the downloaded files. Hint: you should only have to change a small portion of the code for this and each subsequent problem.

6. In the world `harvest1b.json`. Adapt the code so that Reeborg will plant carrots in the empty spots.
7. In the world `harvest1c.json`, Reeborg is carrying an infinite supply of carrots. Adapt the code so that Reeborg plants carrots in the gaps in the field.
8. In the world `harvest5.json`, Reeborg's goal is to pick all of the flowers (daisies and tulips).
9. In the world `harvest7.json`, Reeborg should pick everything except for the smiley tokens (Reeborg knows these as "token").
10. In the world `harvest8.json`, Reeborg should pick everything except the flowers (daisies and tulips).

Prompt: What is a benefit of being able to use multi-branch conditionals over simple "if" or "if..else.." statements? Give an example of how using a multi-branch statement may make your code more DRY.